Some Basic Probability

Continuous Random Variables

- Probability Density Function (p.d.f.)
- Cumulative Distribution Function (d.f.)
- Expected Value

Outline

Last Week

I. Recap - Week 2

II. Plotting (Week 2 Cont.)

Descriptive Statistics

I. Measures of Central Tendency: Mean, Median, Trimmed Mean, Mode

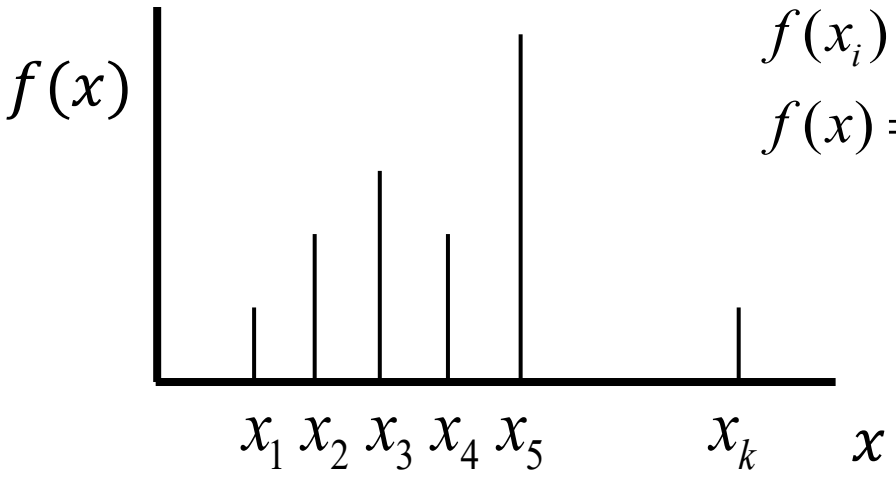
II. Measures of Variability: Range, Interquartile Range, Variance, Standard Deviation

III. Skewness, Kurtosis

IV. Descriptive Statistics Functions
- summary()
- describe()
- describeBy()

Discrete Probability Function (p.f.) of a Random Variable *X*

$$f(x) = \text{Prob}(X = x)$$



$f(x_i)$ $x_1, x_2, \ldots, x_k$

$f(x) = 0$ elsewhere

Properties

$$f(x_i) \geq 0$$

$$\sum_{i=1}^{k} f(x_i) = 1$$

Expected Value of a Function of *X*

$$E[h(X)] = \sum_{x \in D} h(x) f(x)$$

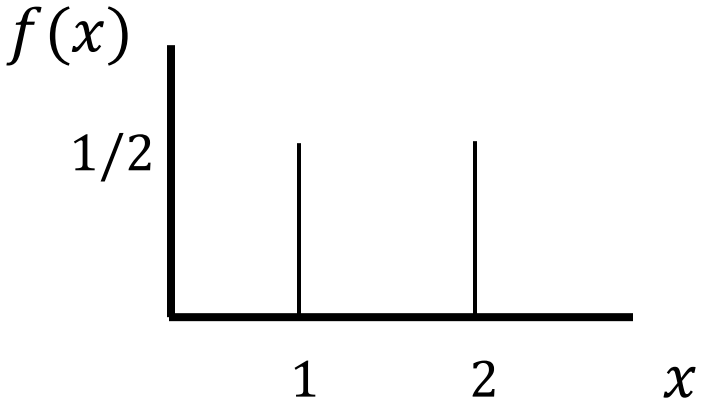
Mean and Variance of *X*

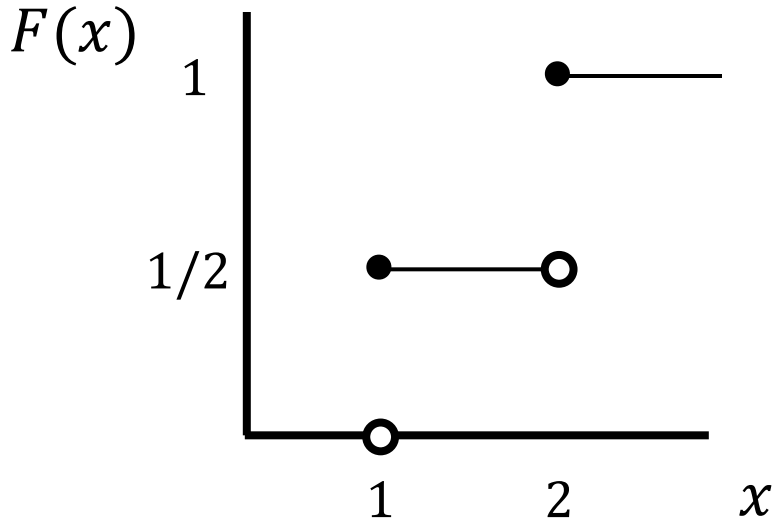
$$E[X] \equiv \mu$$

$$E\left[\left(X - E[X]\right)^2\right] \equiv \sigma^2$$

Distribution Function (d.f.)

$$f(x) = \text{Prob}(X = x) \qquad F(x) = \text{Prob}(X \le x) = \sum_{y:y \le x} f(y)$$





Others: e.g., d.f. for loaded die RV $f(x) = 0.20, \quad x = 1$

$f(x) = 0.16, \quad x = 2, \ldots, 6$

$0, \quad$ elsewhere

Continuous Random Variable

Definition: RV that can take on any value within a range

Example 1: Body temperature of humans:

X in ^{0}C: min $\rightarrow 37^0 \rightarrow$ max

Example 2: Detectable HIV1 virus in children born to infected mothers:

X in viral RNA copies/mL blood: $0 \rightarrow$ max

Example 3: Forced expiratory volume in 1 sec (FEV1):

X in L: min $\rightarrow$ 3.5L $\rightarrow$ max

Degree of Randomness – Probability that *X* is within a range

Probability Density Function (Continuous RV)

How to describe the "degree of randomness" of a continuous RV X?

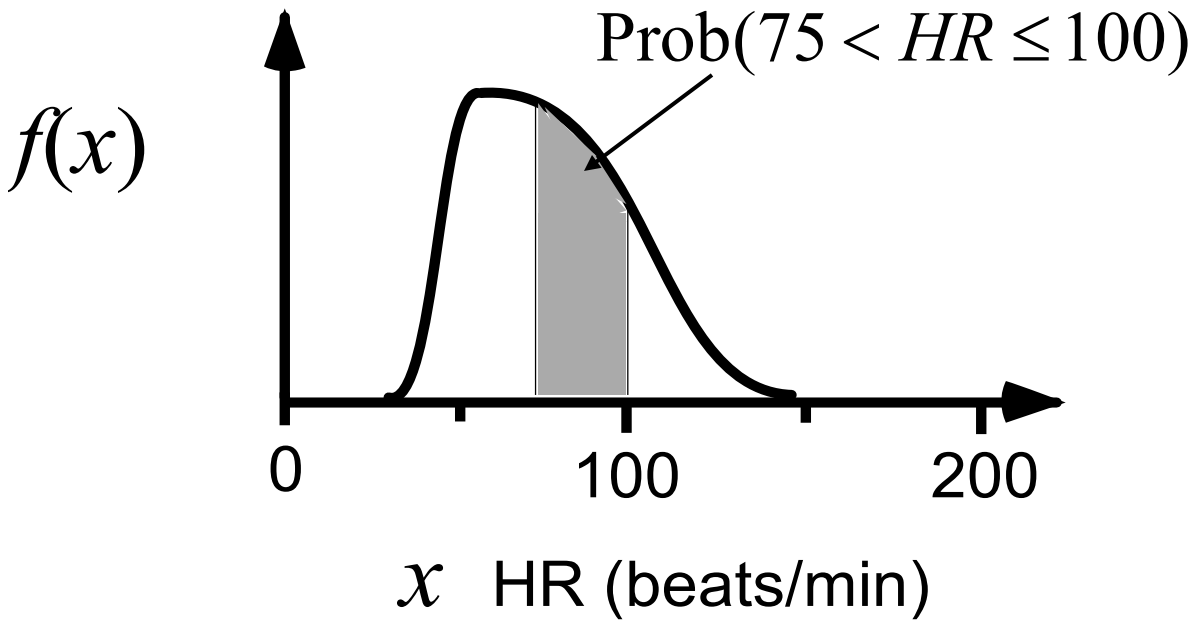
Probability density function (p.d.f.): $f(x)$

$$\text{Prob}(a < X \le b) = \int_a^b f(x)dx$$

Properties

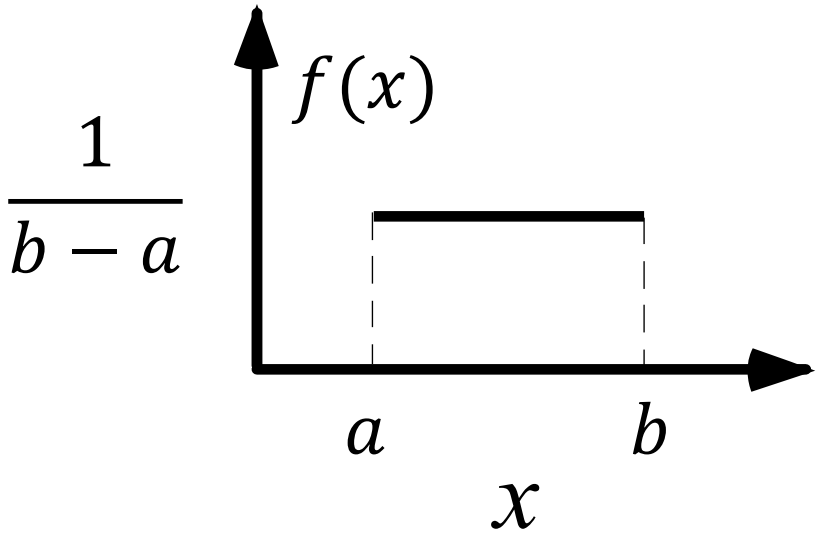
$$f(x) \ge 0$$

$$\int_{-\infty}^{\infty} f(x)dx = 1$$



1. Uniform p.d.f. (Uniform Density)

$$f(x) = \frac{1}{b-a}, \quad a < x \le b, \quad 0 \text{ elsewhere}$$

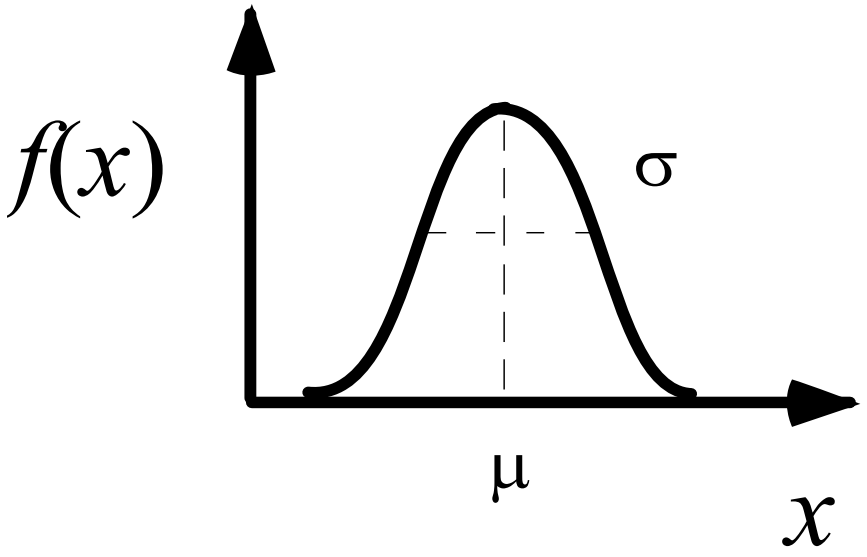


$$X \sim U(a, b)$$

Special case $X \sim U(0, 1)$ (Standard Uniform Density)

2. Normal or Gaussian Density

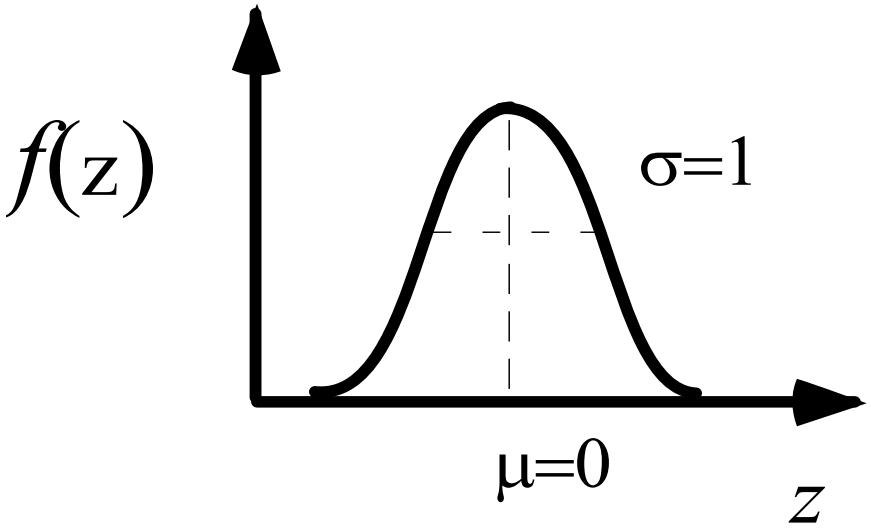
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2} \right)$$



$$X \sim N(\mu, \sigma^2)$$

3. Standard Normal Density (special case) $\mu = 0 \quad \sigma^2 = 1$

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-(z)^2}{2}\right) \qquad \mu = 0 \quad \sigma^2 = 1$$



$$Z \sim N(0, 1)$$

Distribution Function (d.f.)

Let RV $X \sim f(x)$ probability density function

$F(x)$ distribution function

$$F(x) = \text{Prob}(X \le x) = \Pr(X \le x)$$

$$\boldsymbol{F(x) = \int_{-\infty}^{x} f(u) \cdot du}$$

Relationship between p.d.f. and d.f.

Properties

$0 \le F(x) \le 1 \quad -\infty < x < \infty$

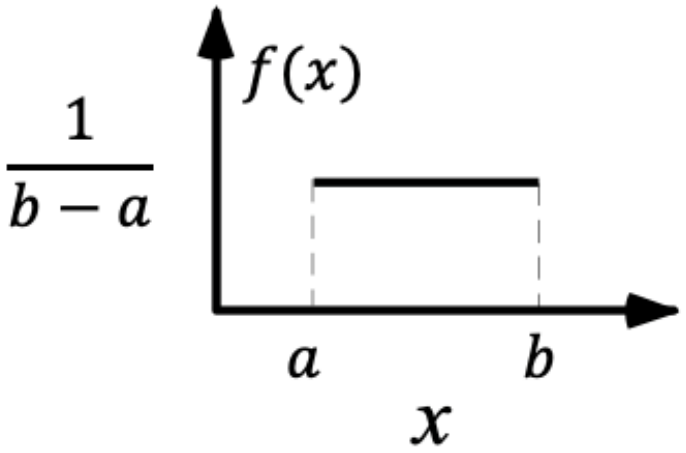
$F(-\infty) = 0 \quad F(\infty) = 1$

$F(x)$ nondecreasing with increasing x

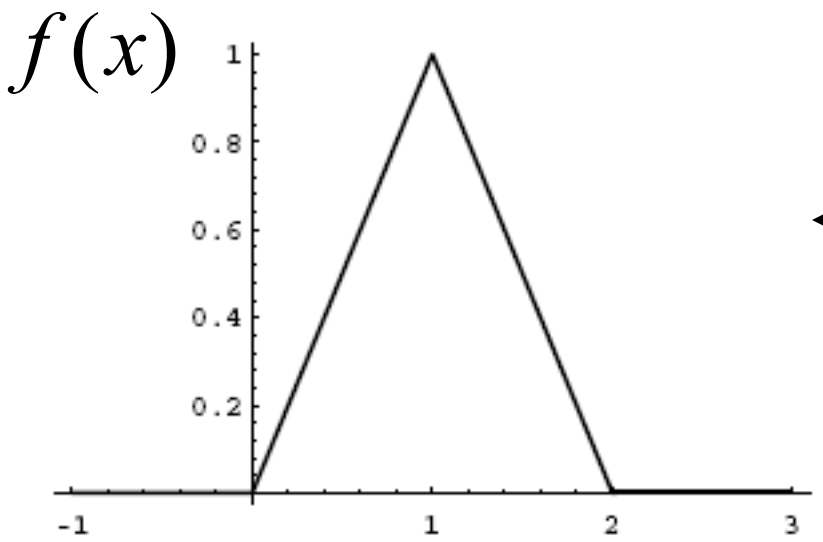
Distribution Function (d.f.)

Examples

p.d.f. **d.f.**



$\longleftrightarrow$ $F(x) = ?$



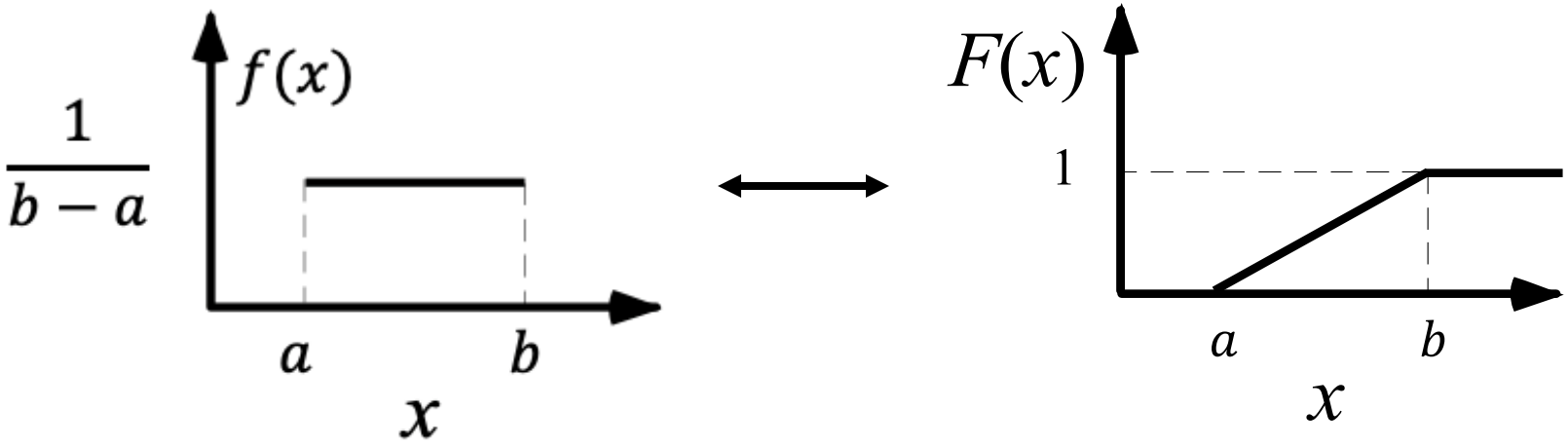
$\longleftrightarrow$ $F(x) = ?$

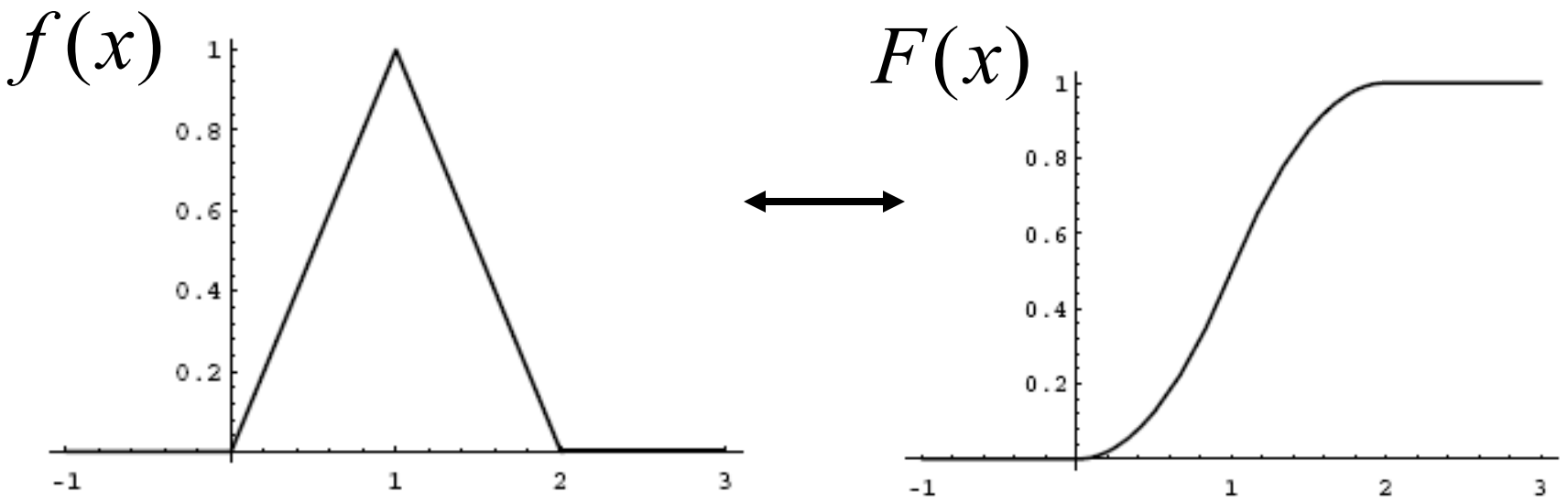
Distribution Function (d.f.)

$$F(x) = \int_{-\infty}^{x} f(u) \cdot du$$

Examples

p.d.f. **d.f.**



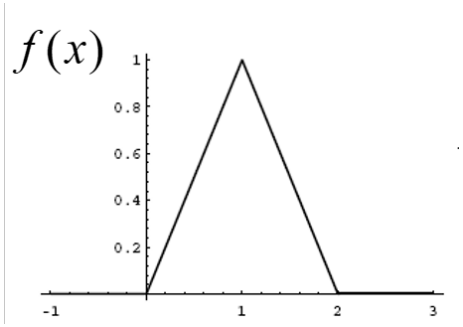


Distribution Function (d.f.)

Example

p.d.f. **d.f.**

$$f(x) = \begin{cases} 0 & x < 0 \\ x & 0 \le x < 1 \\ 2 - x & 1 \le x < 2 \\ 0 & 2 \le x \end{cases} \longleftrightarrow F(x) = ?$$



Other Relationships Involving d.f.

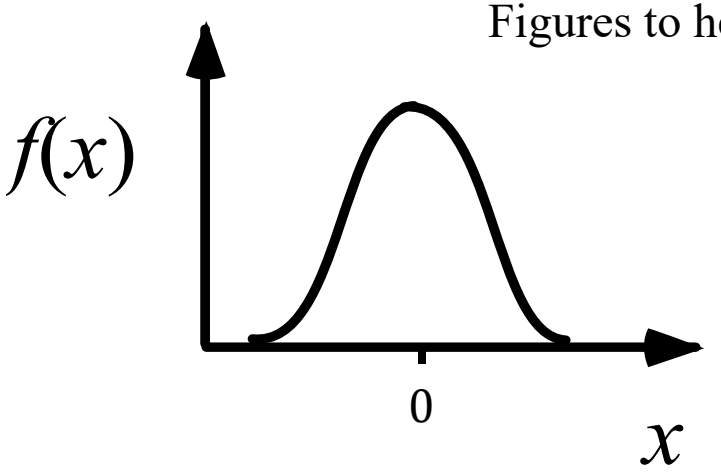
Recall: $F(x) = \Pr(X \le x)$

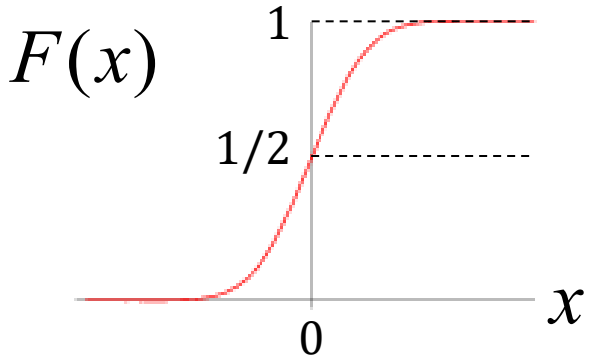
$$\boldsymbol{F(x) = \int_{-\infty}^{x} f(u) \cdot du}$$

$$\Pr(X > x) = 1 - F(x)$$

$$\Pr(x_1 < X \le x_2) = F(x_2) - F(x_1) = \int_{x_1}^{x_2} f(x)\,dx$$

Figures to help illustrate these ideas





Expected Value (Mean) of the RV *X*

Definition:

$$X \sim f(x)$$

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx \equiv \mu$$

Example:

$$f(x) = \frac{1}{b-a}, \quad a < x \le b, \quad 0 \text{ elsewhere}$$

$$E[X] = \int_{a}^{b} x \left(\frac{1}{b-a} \right) dx = \left(\frac{1}{b-a} \right) \frac{x^2}{2} \Big|_a^b = \frac{a+b}{2}$$

Expected Value of a Function of the RV *X*

Let X denote a continuous RV with p.d.f. $f(x)$

Define a function of X as $h(X)$

$$E[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx$$

Example:

New RV Y $\quad Y = a + bX = h(X)$

Find $\quad E[Y] = E[h(X)]$

$$E[Y] = \int_{-\infty}^{\infty} h(x) f(x)\, dx = \int_{-\infty}^{\infty} \left(a + bx\right) f(x)\, dx$$

$$= a\int_{-\infty}^{\infty} f(x)\, dx + b\int_{-\infty}^{\infty} xf(x)\, dx$$

$$E[Y] = a + bE\left[X\right]$$

$$\mu_Y = a + b\mu_X$$

Variance of a Continuous Random Variable

$$E[\left(X - E\left[X\right]\right)^2] = \int_{-\infty}^{\infty} \left(x-\mu\right)^2 f(x)dx \equiv \sigma^2$$

Example:

p.d.f. of X $\quad f(x) = \frac{1}{20}, \quad 20 < x \le 40, \quad 0$ elsewhere

Find variance of X

$$E[\left(X-\mu\right)^2] = \int_{20}^{40} \left(x-\mu\right)^2 \frac{1}{20} dx = ?$$

Exercises using Expected Value

Let X denote a continuous RV with p.d.f. $f(x)$

Exercise 1:

Show: $\sigma^2 \equiv E[(X-\mu)^2] = E[X^2] - \mu^2$

Exercise 2:

Define New RV Y $\quad Y = a + bX = h(X)$

Show: $\sigma_Y^2 = b^2 \sigma_X^2$

Session 4 Lecture Exercise

Let X denote the amount of time it takes for a new analytical chemistry instrument to analyze a patient's blood sample to determine the concentration of cortisol in the sample. Numerous factors influence the analysis time (e.g., actual cortisol concentration in the sample, size of the sample, presence of other hormones, etc.), but the maximum analysis time is 2 hours. Assume the p.d.f. for X is

$$f(x) = 0.5x \qquad 0 \le x \le 2$$
$$= 0 \qquad \text{otherwise}$$

a) Determine the d.f. $F(x)$ and plot it
b) Calculate the probability that the analysis time is < 1 hr
c) Calculate expected analysis time for a random sample
d) From $F(x)$, calculate the probability the analysis time is > 1.5 hr
e) What is the median analysis time?

Session 4 Lecture Exercise

a) Determine the d.f. $F(x)$ and plot it
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